

demo

email | location | linkedin | github --- ## PROFESSIONAL SUMMARY As a skilled Data Scientist with a strong background in Computer Vision and Machine Learning, I leverage my expertise in Python, PyTorch, and OpenCV to develop innovative solutions for real-world problems. With a proven track record of delivering high-impact projects, I am confident in my ability to drive business growth through data-driven insights. My experience in managing the machine learning lifecycle, from labeling to model training, deployment, monitoring, scaling, and iteration, makes me a strong fit for this role. I am excited about the opportunity to work with Lineage Logistics and contribute to the development of cutting-edge computer vision technologies. --- ## TECHNICAL SKILLS - ML Frameworks: PyTorch, Scikit-learn

- Programming Languages: Python, CUDA C++
- Computer Vision: OpenCV, Transformers
- Data Science: NumPy, Pandas, Jupyter Notebook
- Operating Systems: Windows, Linux --- ## WORK EXPERIENCE #### Undergraduate Researcher – MedOptm Project @ Gujarat Technological University (GTU) — Prof. Raj Hakani | Oct. 2025 – Present
- Developed a compact vision-language model integrating CvT, Token Packer, and LLaMA (1.7B) for medical visual question answering, achieving a METEOR score of 0.33.
- Designed a scalable dataset pipeline using biomedical caption datasets, enabling efficient training and inference.
- Optimized model efficiency for real-world clinical usability, ensuring seamless integration with downstream decision-making systems. #### Research Intern @ Carnegie Mellon University — Xu Lab | Feb. 2026 – Present
- Developed deep learning models for brain MRI surface registration, improving anatomical alignment and preserving geometric consistency across datasets.
- Designed end-to-end pipelines for learning-based anatomical alignment, ensuring spatial structure preservation in medical imaging tasks.
- Utilized Python and PyTorch to implement and train models, demonstrating strong programming ability. #### Machine Learning Intern @ Infosys Springboard | Nov. 2025 – Jan. 2026
- Led development of ANPR and ATCC systems, building deep learning models for multi-class vehicle detection and classification.
- Engineered automated data pipelines and applied data augmentation techniques, improving model robustness and scalability.
- Achieved 85% accuracy across 12 vehicle classes and improved overall performance by 4%. #### Undergraduate Researcher – Brain Tumor Classification @ Gujarat Technological University (GTU) — Prof. Raj Hakani | Jun. 2025 – Aug. 2025
- Designed a CNN-based MRI tumor classification model, improving generalization across multi-class datasets.
- Implemented a two-stage training strategy (neural head + fine-tuning) using a VGG16-based architecture.
- Achieved 97.88% classification accuracy, with work accepted at ICMLDE 2025. --- ## PROJECTS #### Brain-MRI-Surface-Registration

Tech Stack: HTML, PyTorch, Graph Neural Networks

- Developed a complete PyTorch implementation of an end-to-end deep learning pipeline for 3D brain surface reconstruction.
- Utilized Vision Transformers and Graph Neural Networks to improve model performance and efficiency.
- Demonstrated strong understanding of Deep Learning concepts and Computer Vision techniques. #### Image-Classification-Medical-

Tech Stack: Jupyter Notebook, Python, Scikit-learn

- Developed a solution for image classification using Python and Scikit-learn.
- Implemented data preprocessing and feature engineering techniques to improve model

performance.

- Achieved high accuracy in image classification tasks, demonstrating strong skills in Machine Learning. ### Emotion-Prediction-App

Tech Stack: HTML, Python, PyTorch

- Built a sentence-based emotion prediction web app using Python and PyTorch.
- Utilized Transformers and Deep Learning techniques to improve model performance and efficiency.
- Demonstrated strong understanding of Natural Language Processing concepts and Computer Vision techniques. --- ## EDUCATION Bachelor of Engineering in Computer Science at Gujarat Technological University (2027) | CGPA: 9.35 --- ## ACHIEVEMENTS & CERTIFICATIONS - Introduction to Machine Learning: IIT Madras
- Database Management Systems: IIT Kharagpur
- First author of a research paper published in Elsevier Procedia Computer Science on brain tumor classification using transfer learning with pre-trained CNNs, achieving 98.77% accuracy across four MRI tumor categories.
- Selected for Amazon ML Summer School 2026, ranking among the top ~2.3% of applicants.
- Published a Hugging Face dataset that surpassed 4,000+ downloads within its first month of release.
- Winner, GTU State-Level Hackathon.